



Redes Neuronales Convolucionales (I)

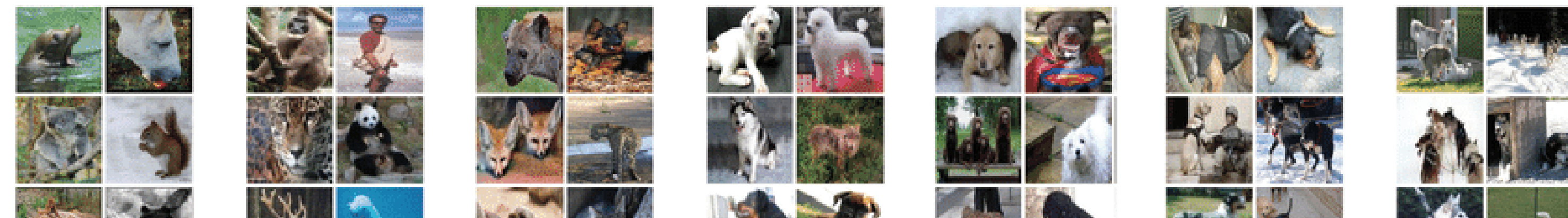
Redes Neuronales y Aprendizaje Profundo

01

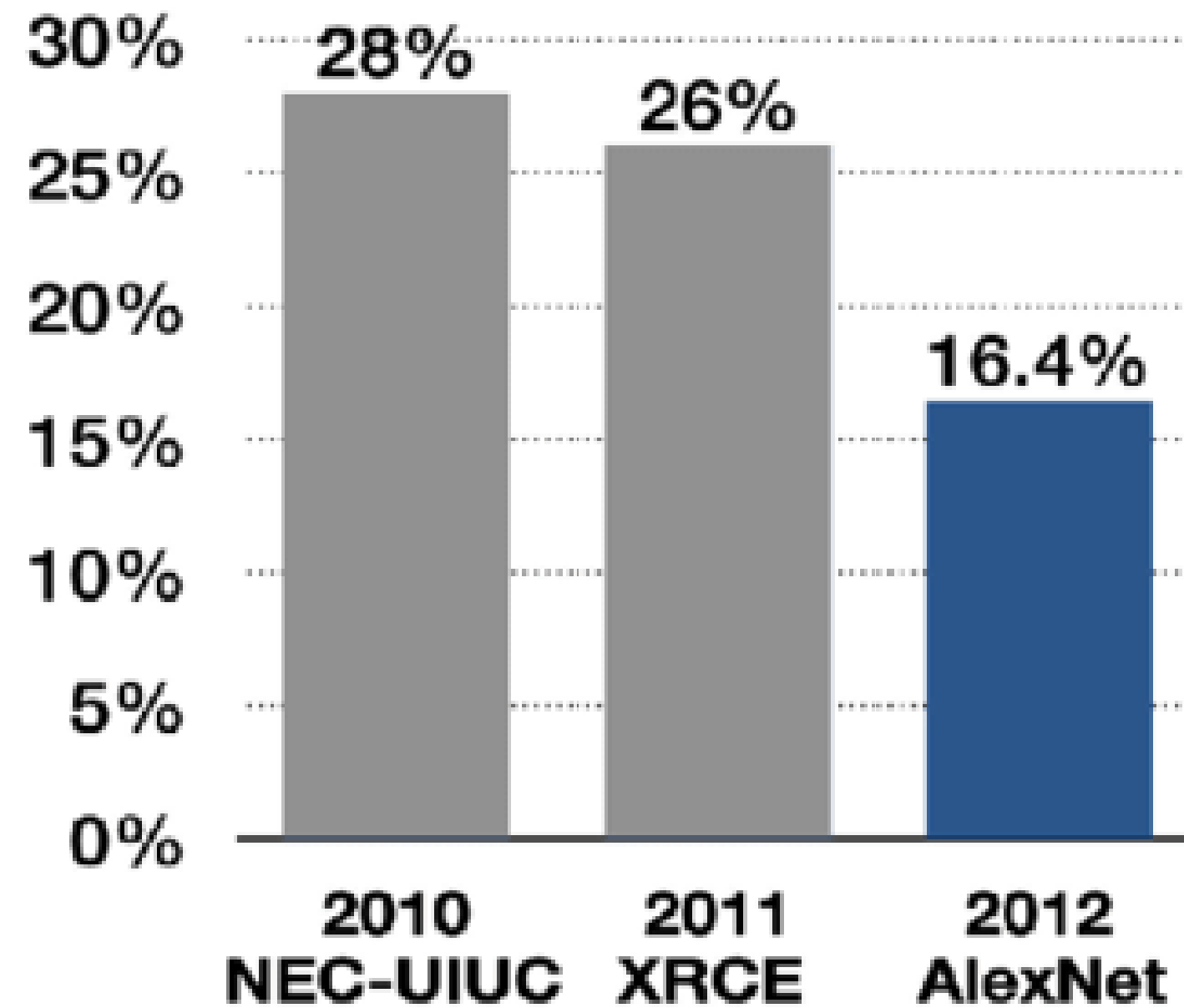
Introducción

Imagenet





Top-5 error



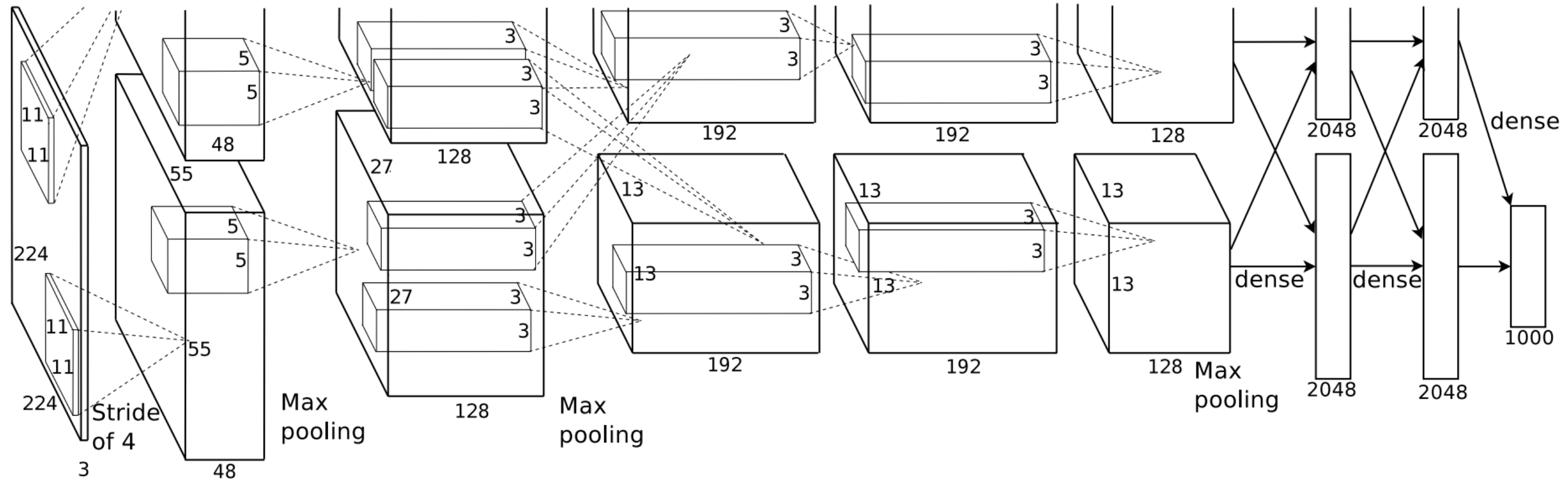
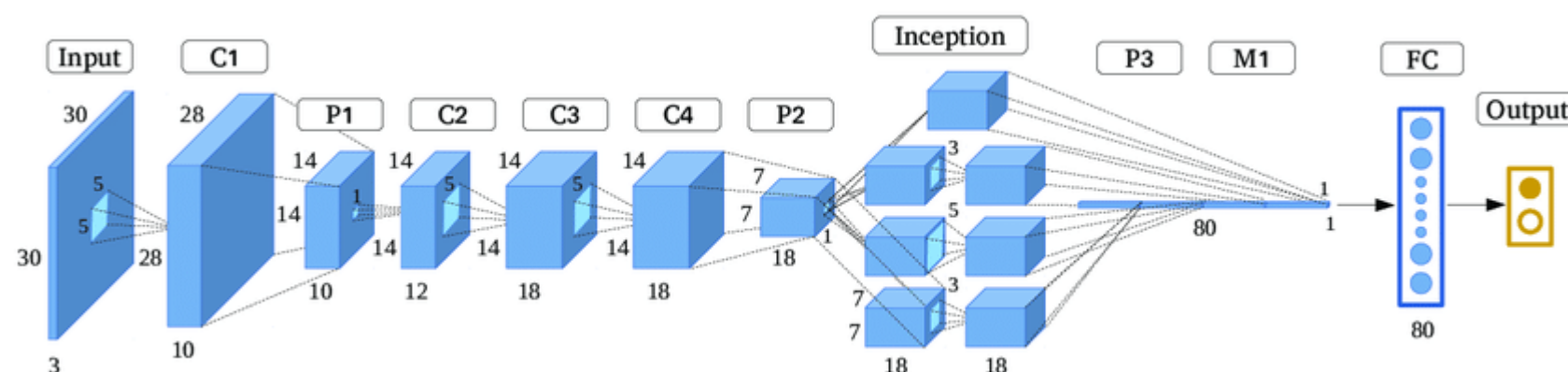
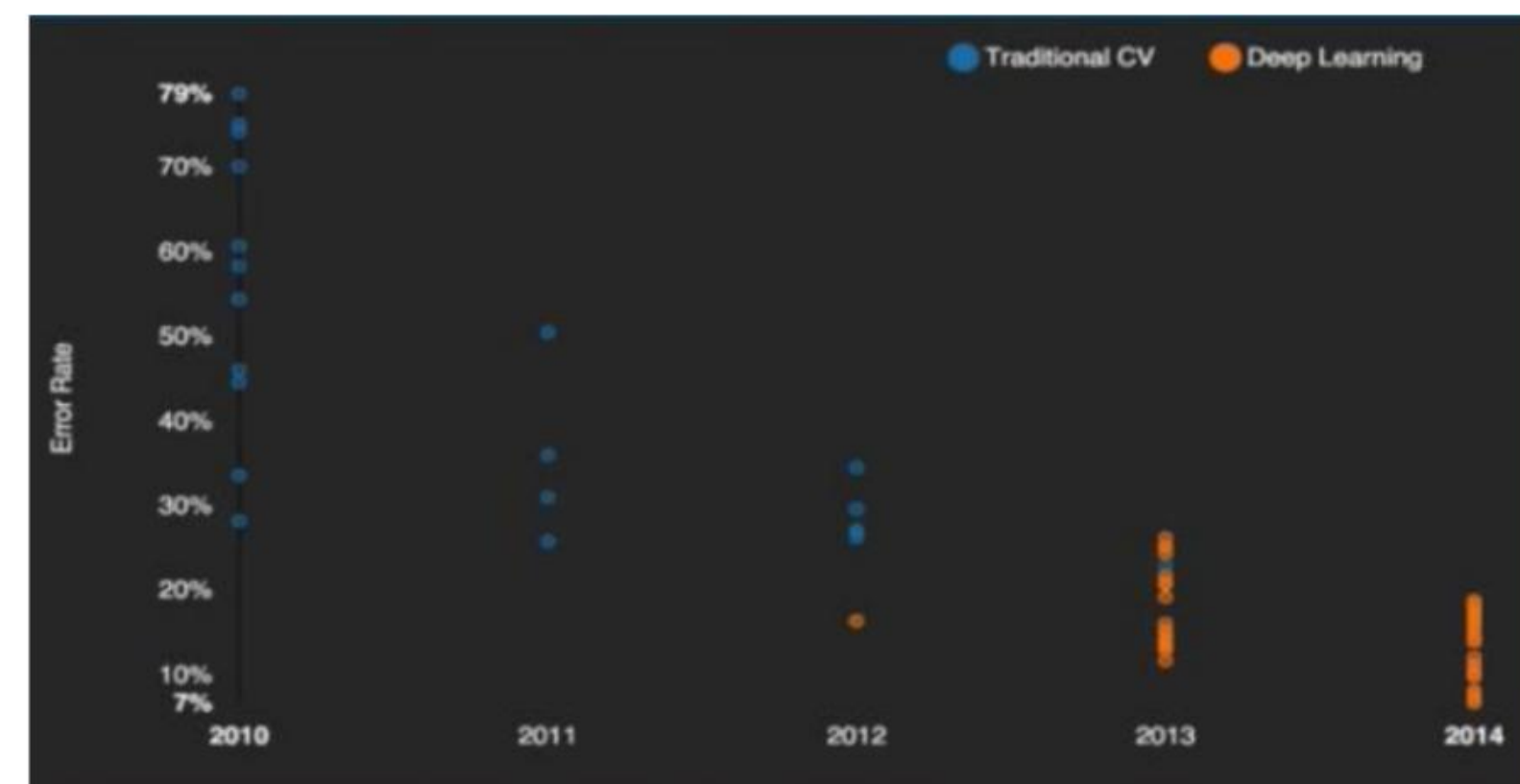
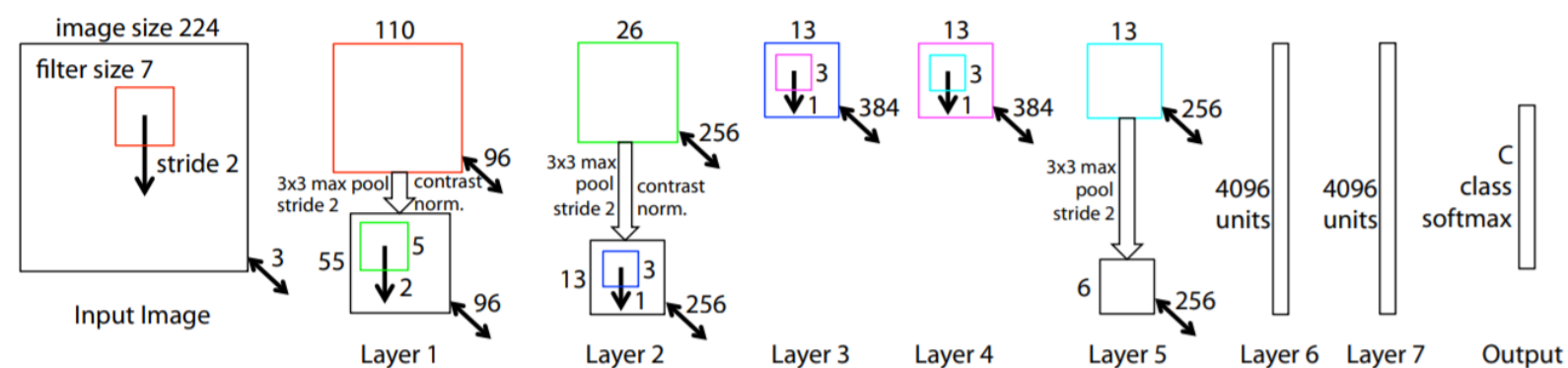
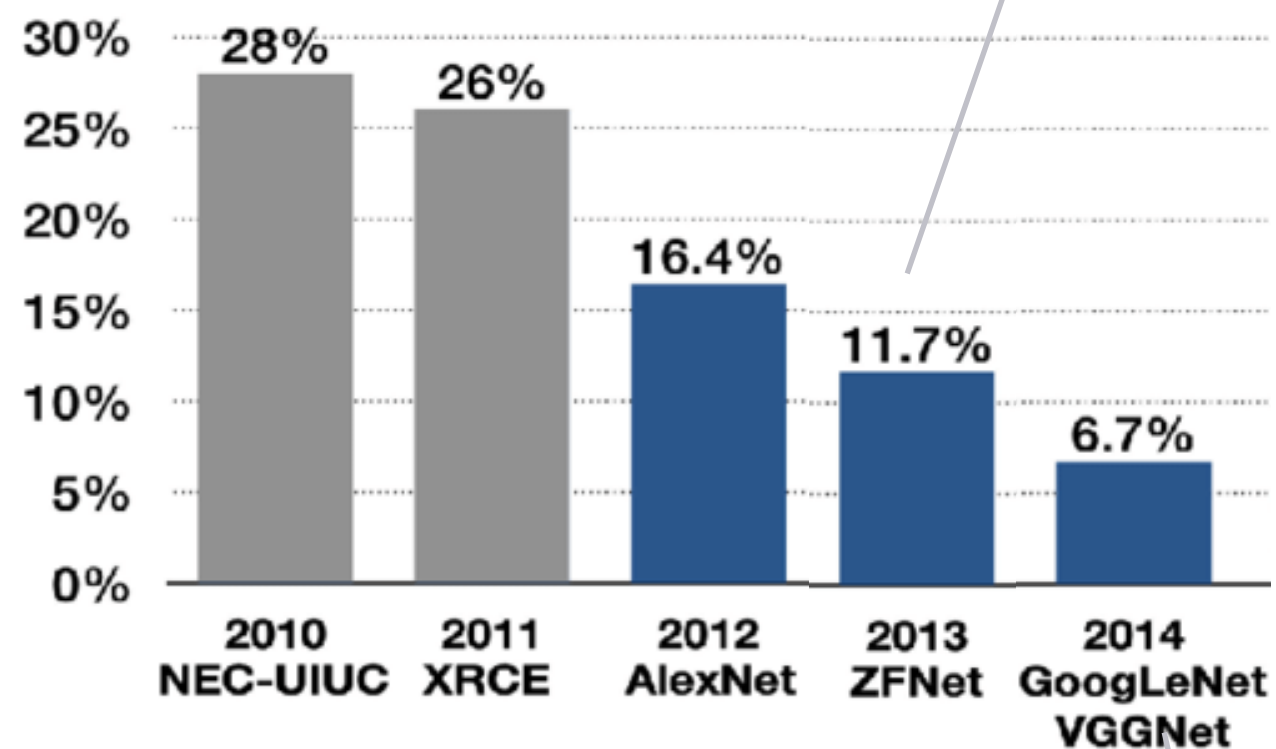


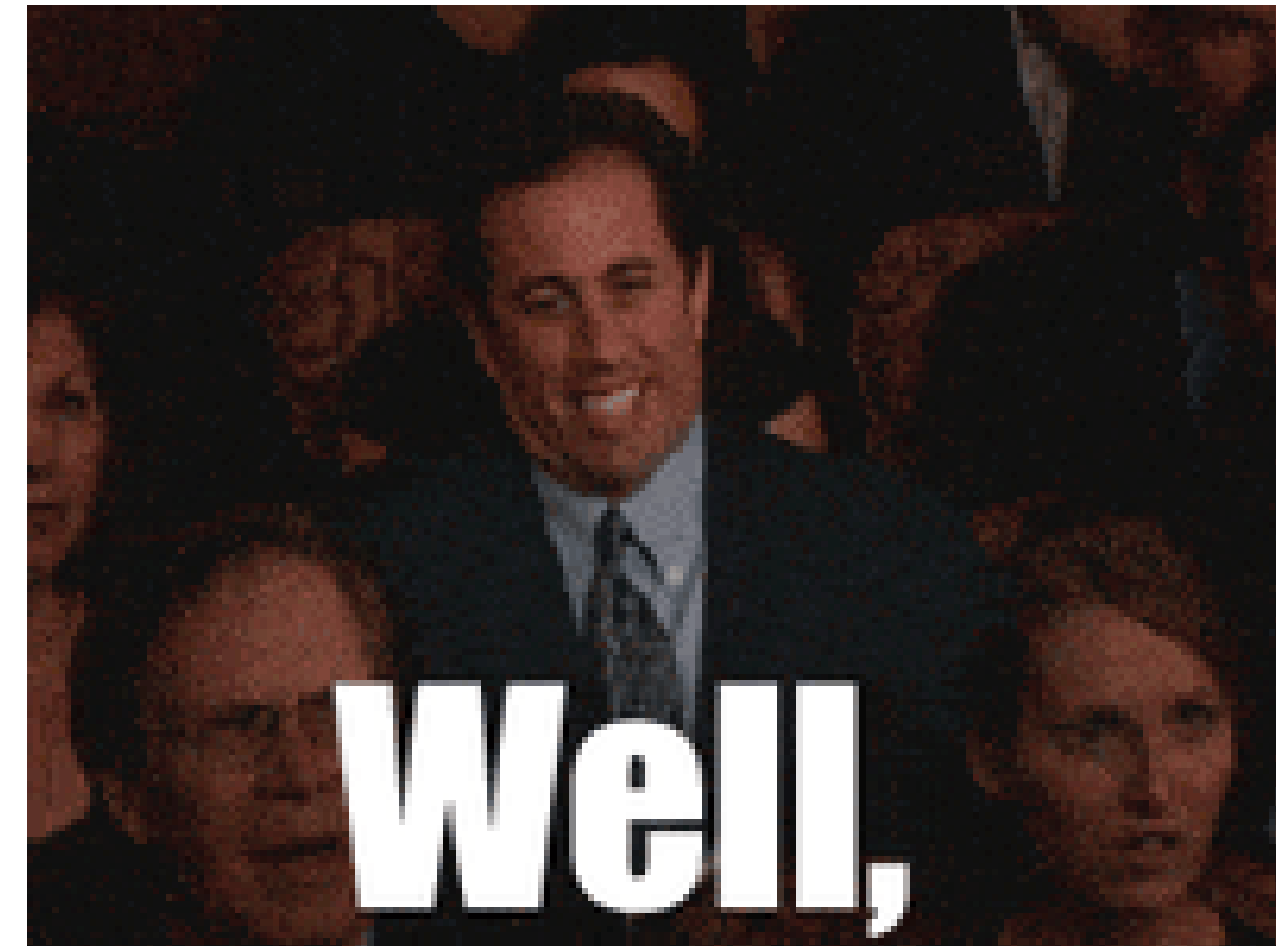
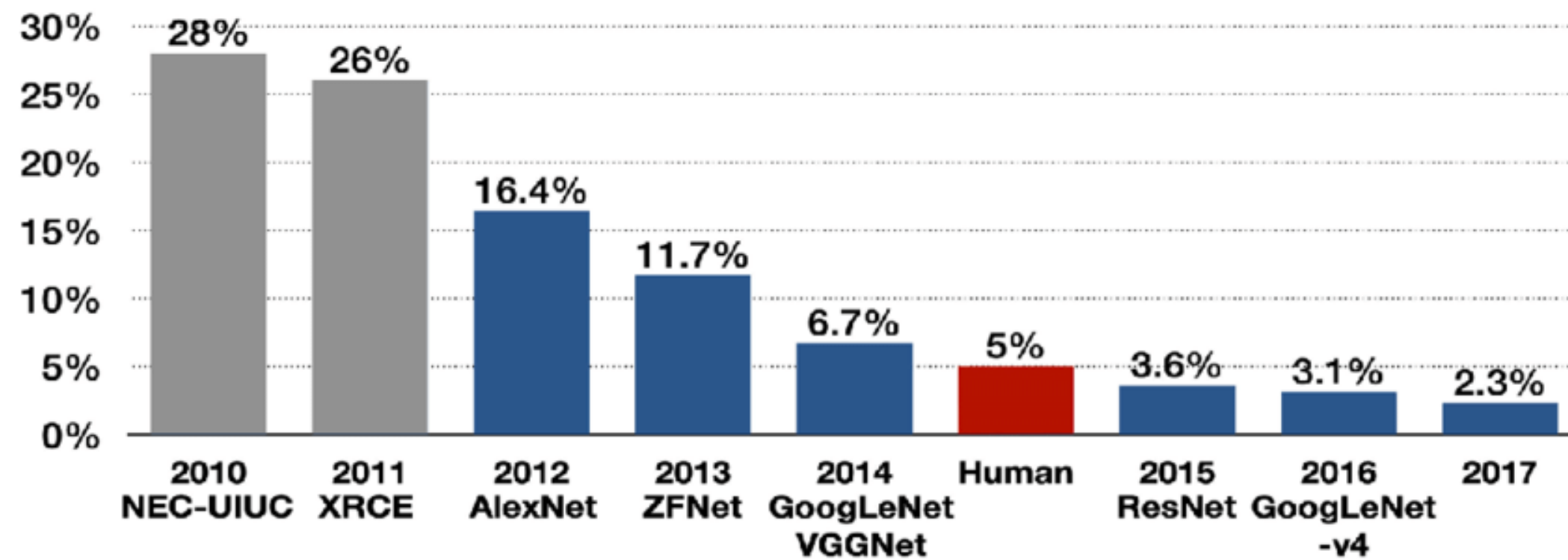
Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

ILSVRC

Top-5 error

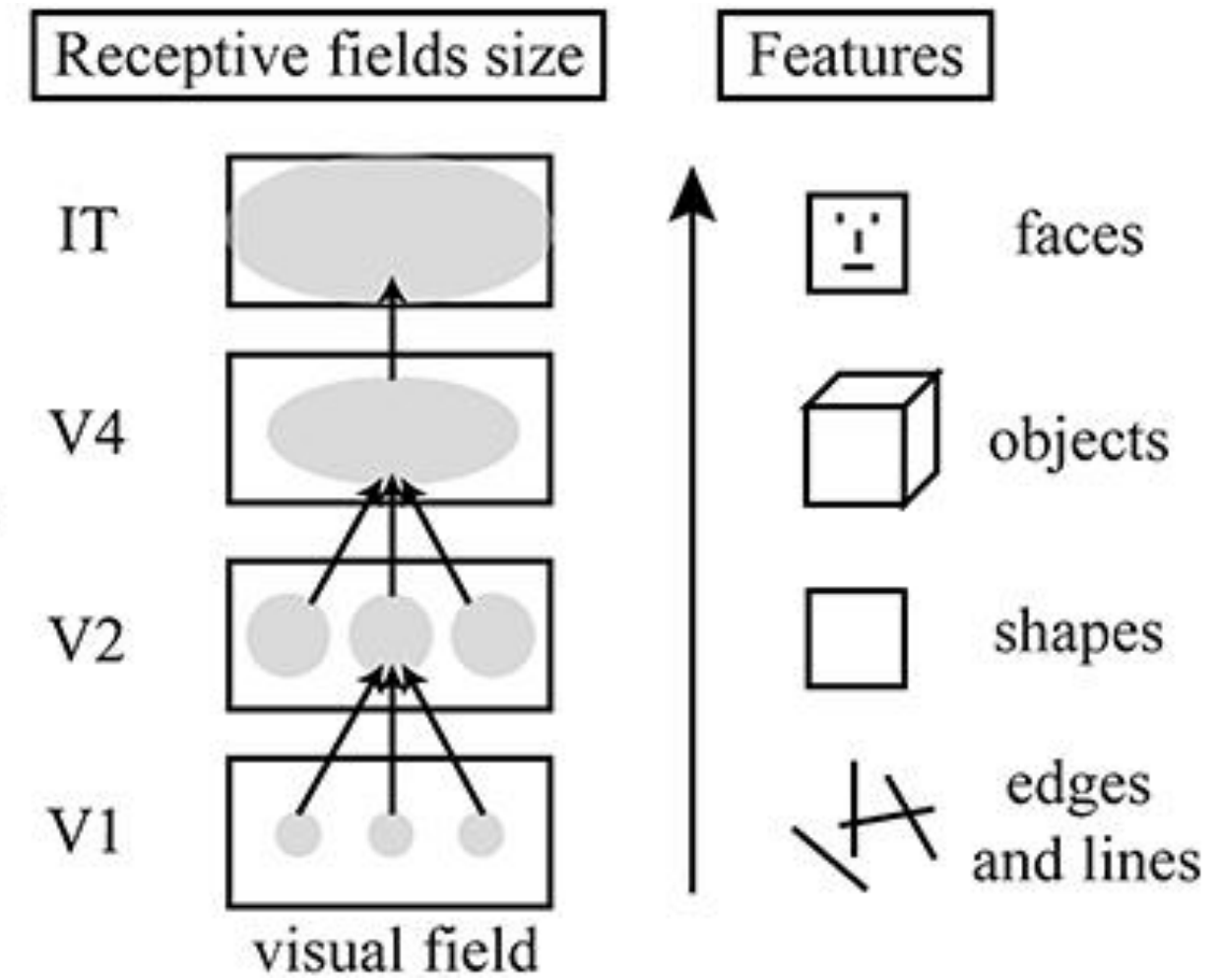
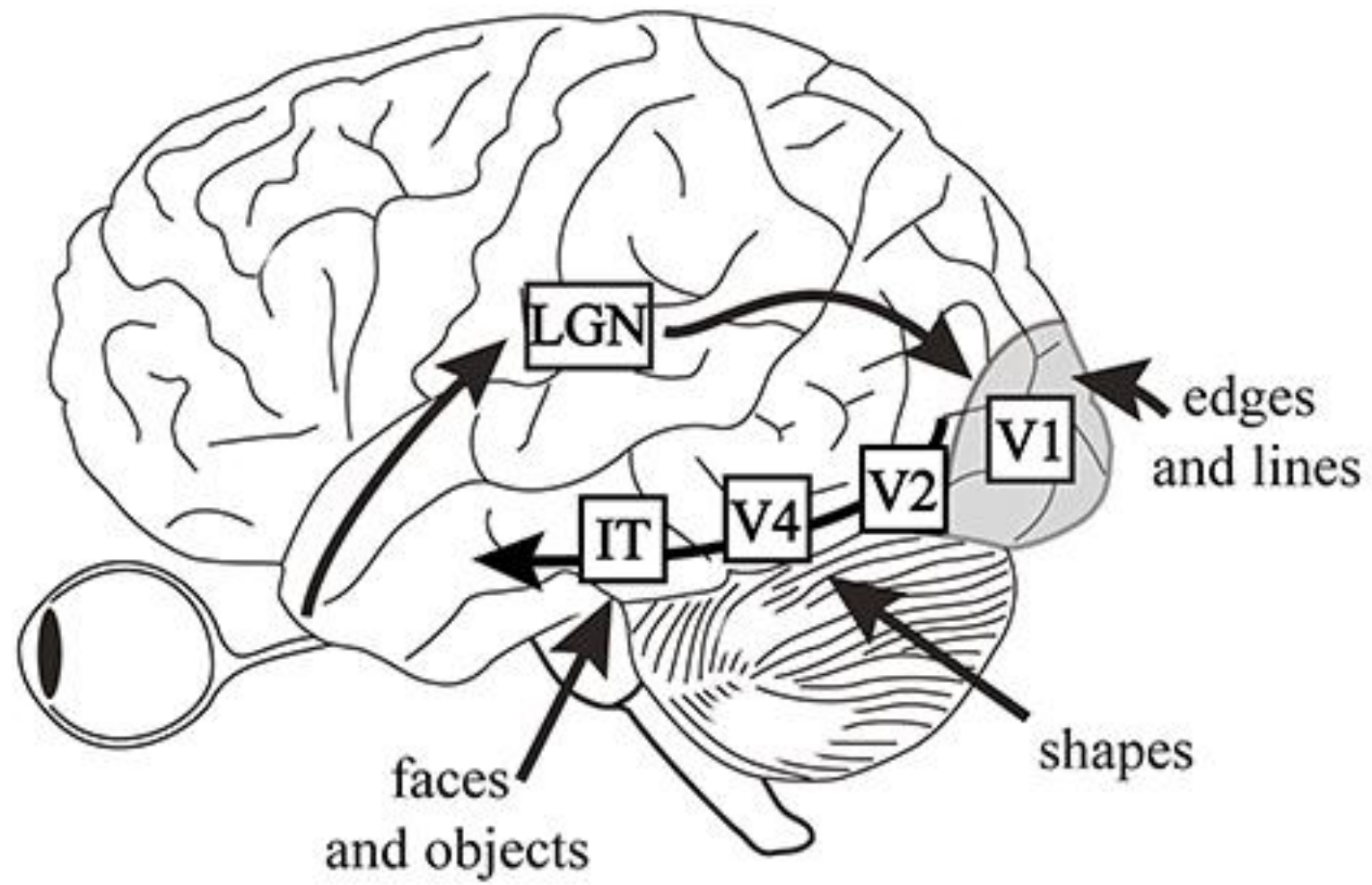


Top-5 error

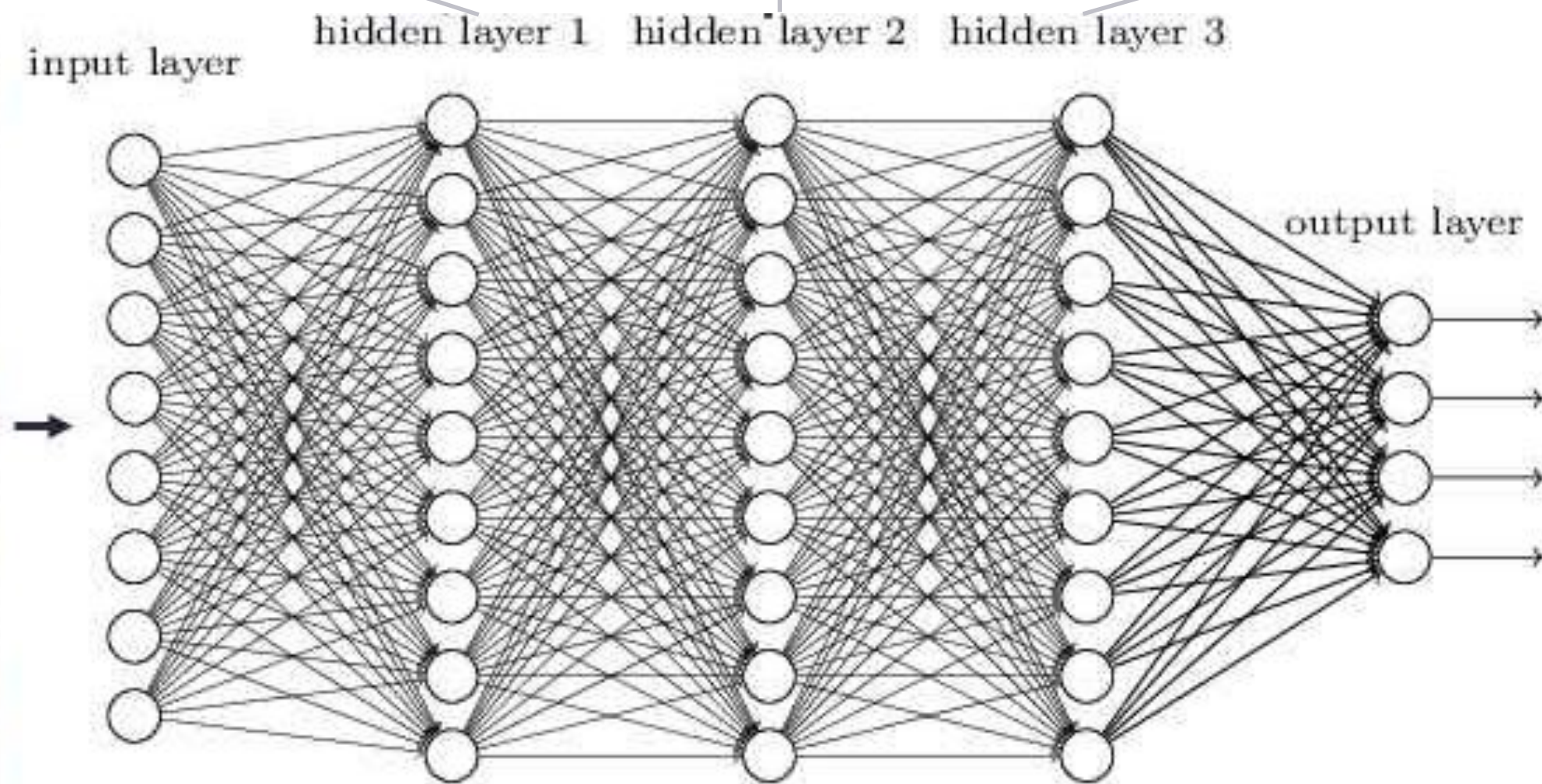
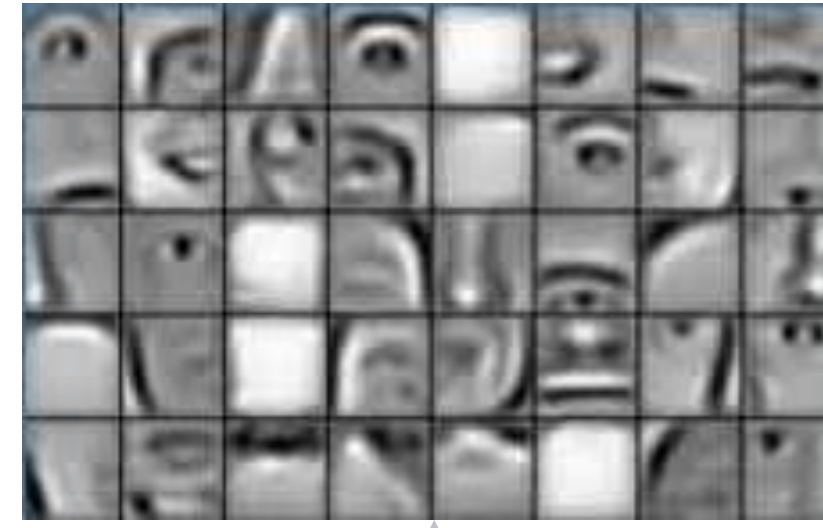
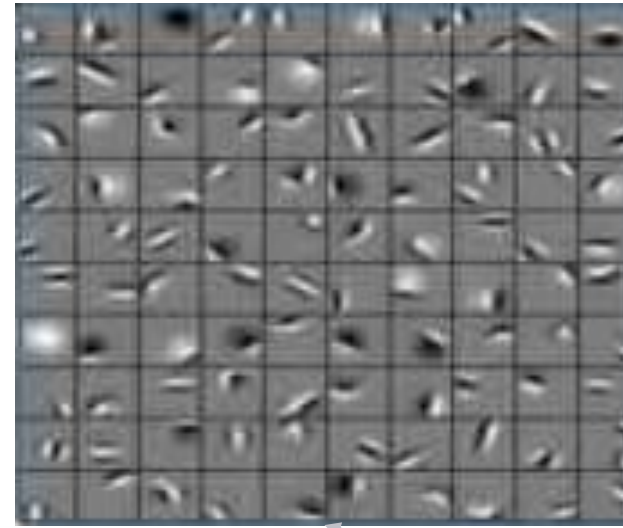




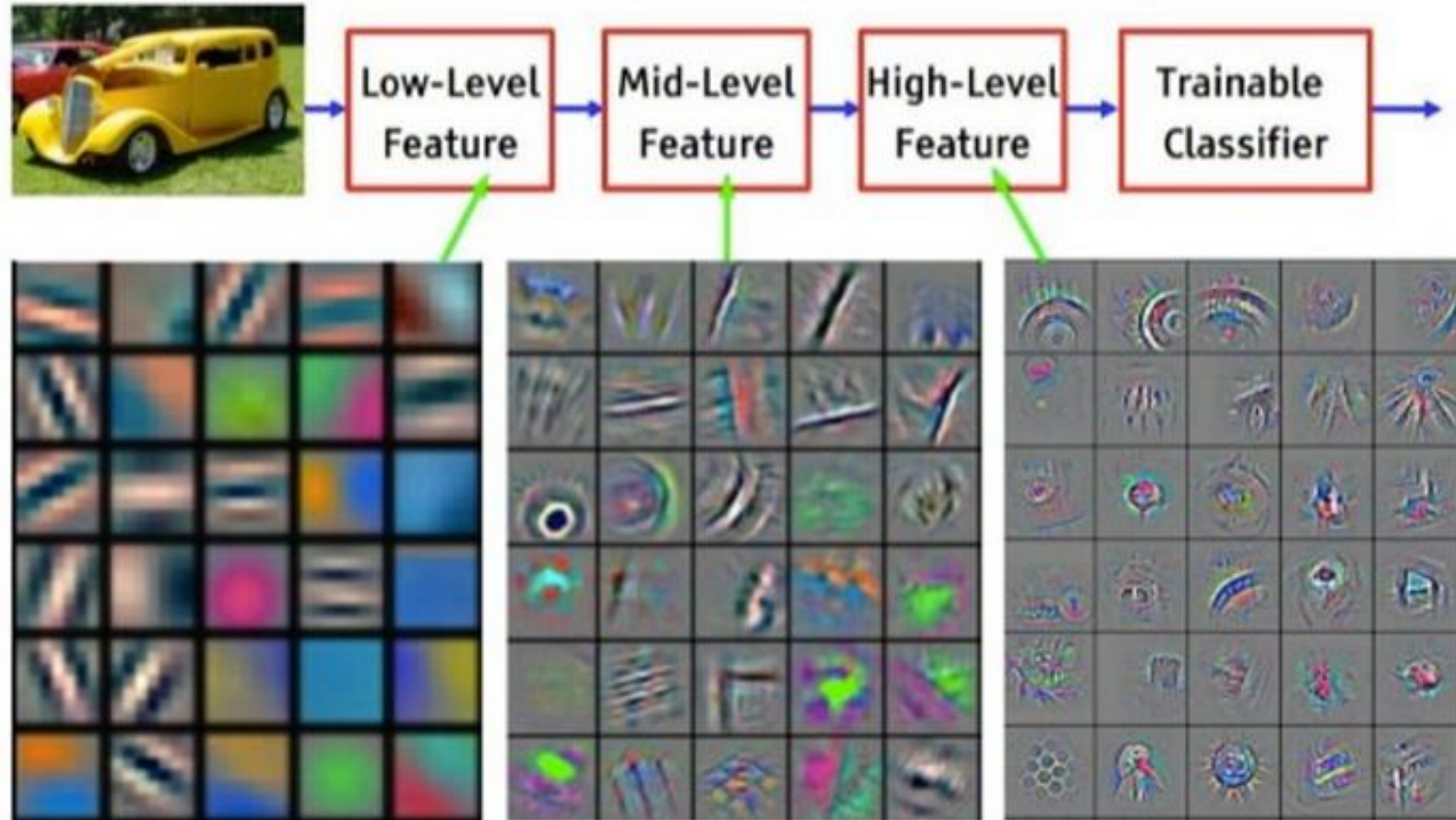
El Cortex Visual



Redes convolucionales



Convolutional Neural Network

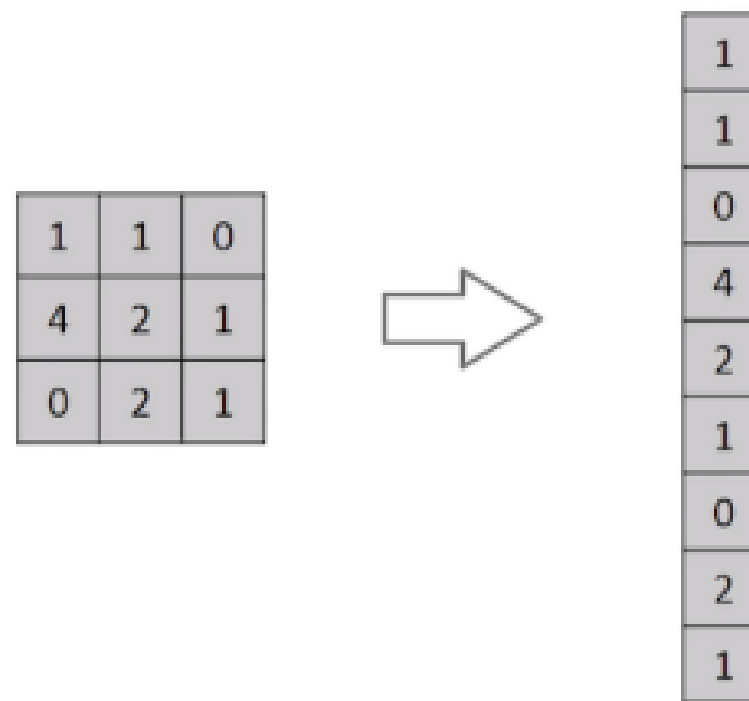


02

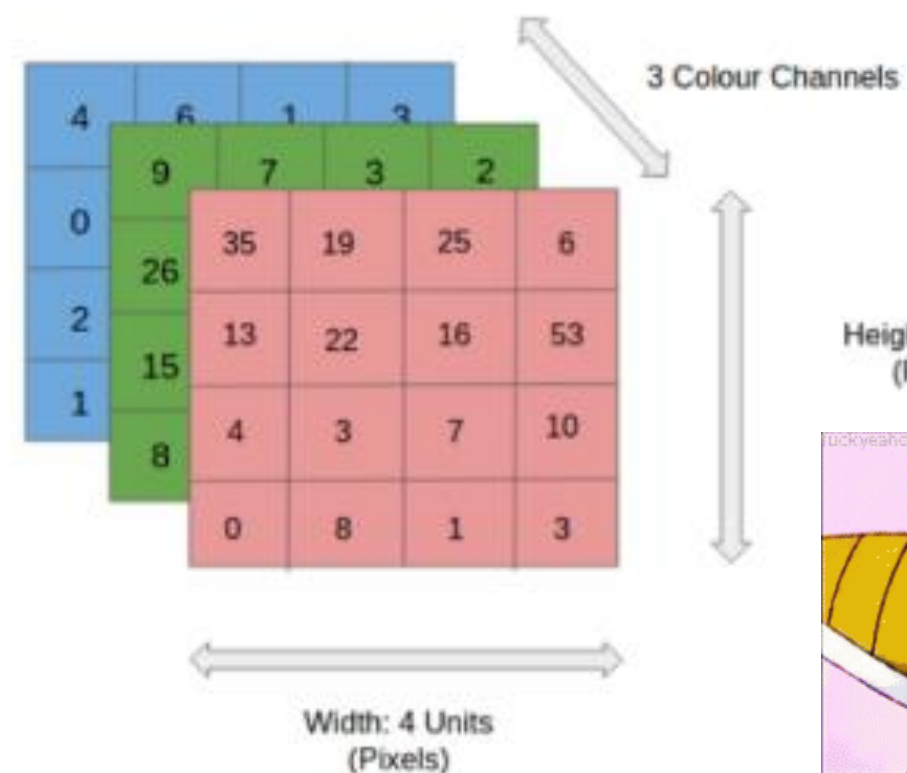
Problemas con Imágenes

3	4	2	1	9	5	6	2	1	8
8	9	1	2	5	0	0	6	6	4
6	7	0	1	6	3	6	3	7	0
3	7	7	9	4	6	6	1	8	2
2	9	3	4	3	9	8	7	2	5
1	5	9	8	3	6	5	7	2	3
9	3	1	9	1	5	8	0	8	4
5	6	2	6	8	5	8	8	9	9
3	7	7	0	9	4	8	5	4	3
7	9	6	4	7	0	6	9	2	3

Problemas con imágenes



- Problema computacional



Problemas con imágenes

More than 270,000 parameters?

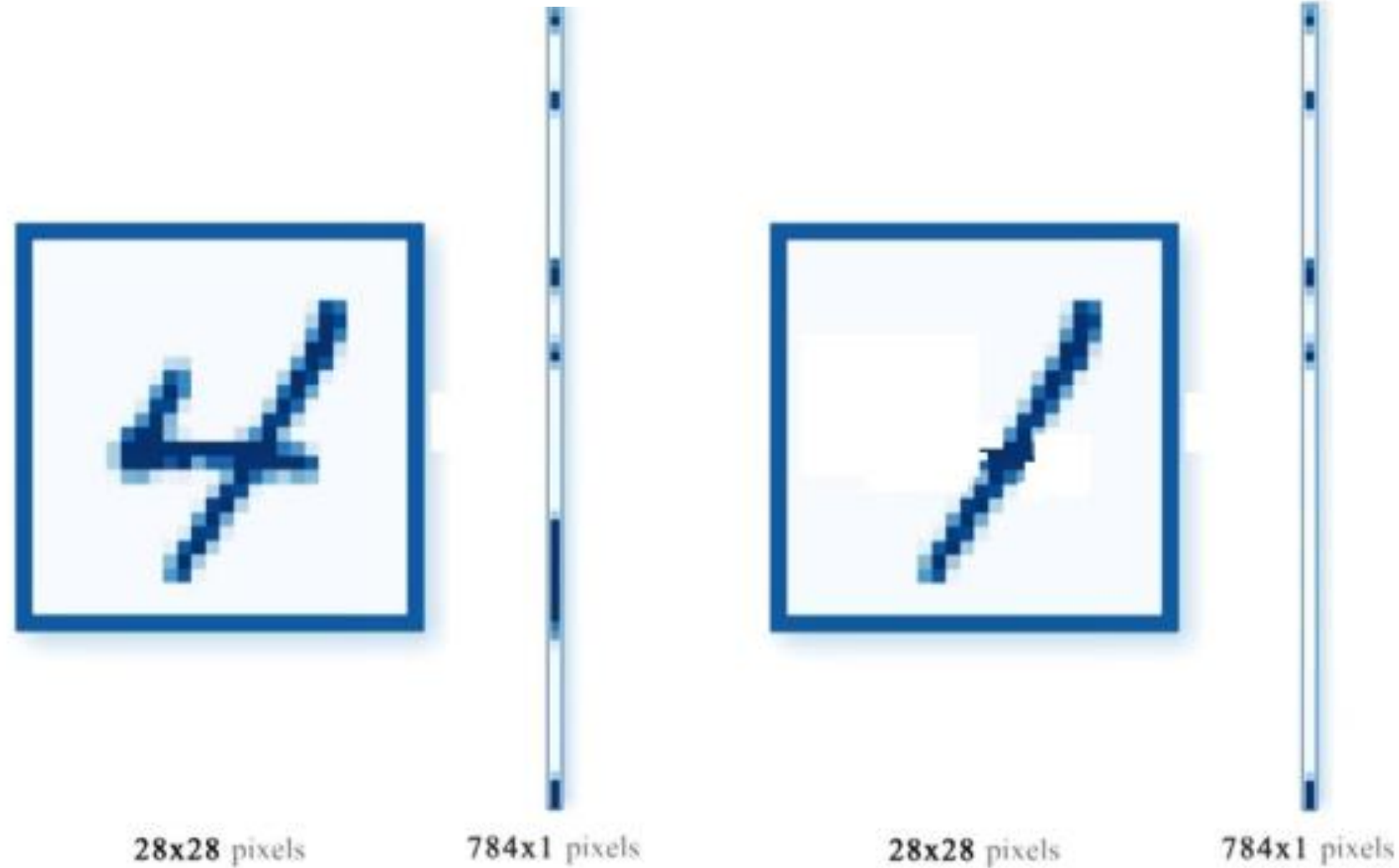


- Problema computacional
- Problema de aprendizaje

Problemas con imágenes



Problemas con imágenes



- Problema computacional
- Problema de aprendizaje
- Problema espacial

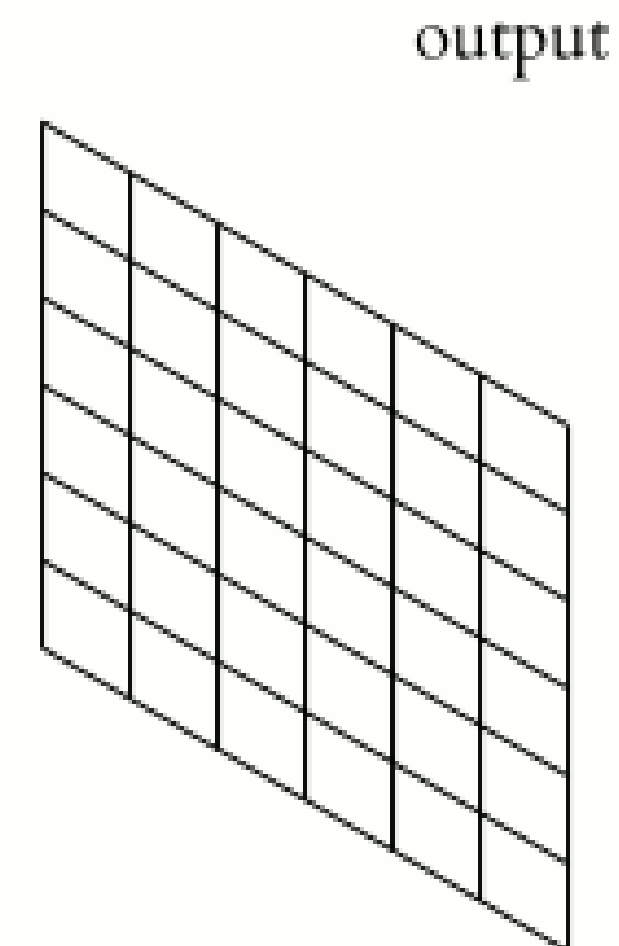
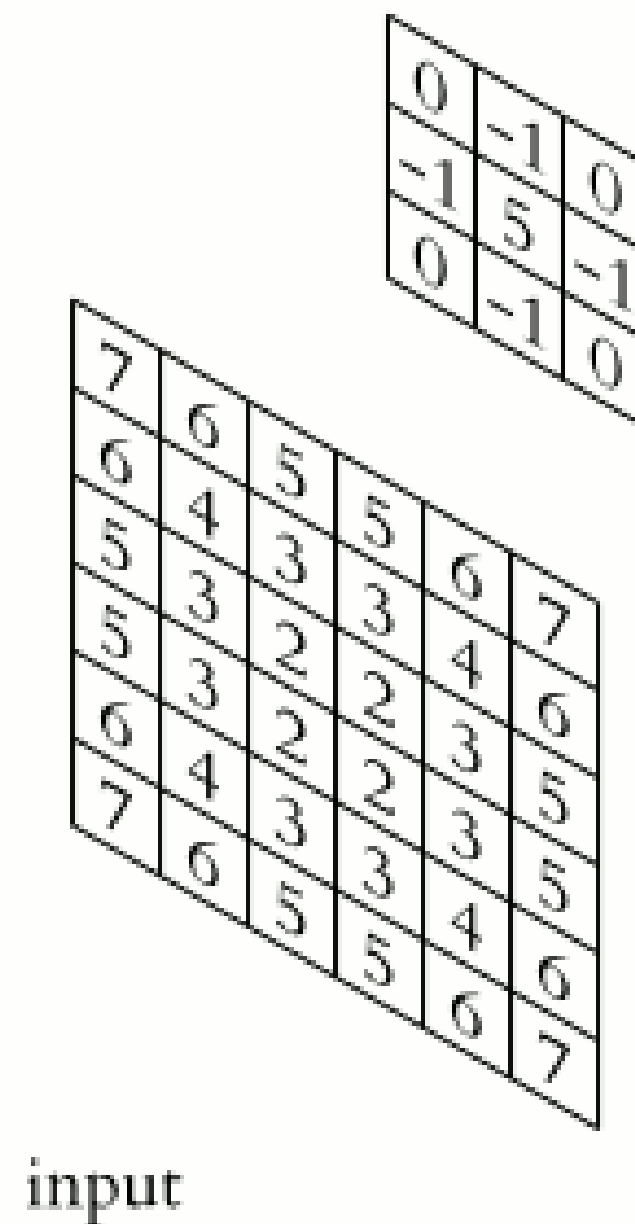
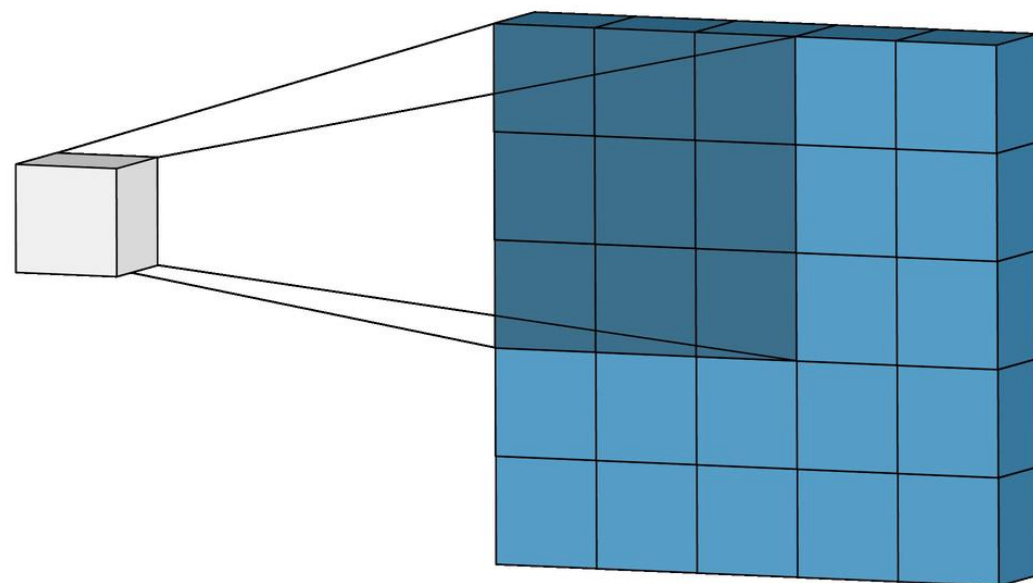
03

Redes Convolucionales

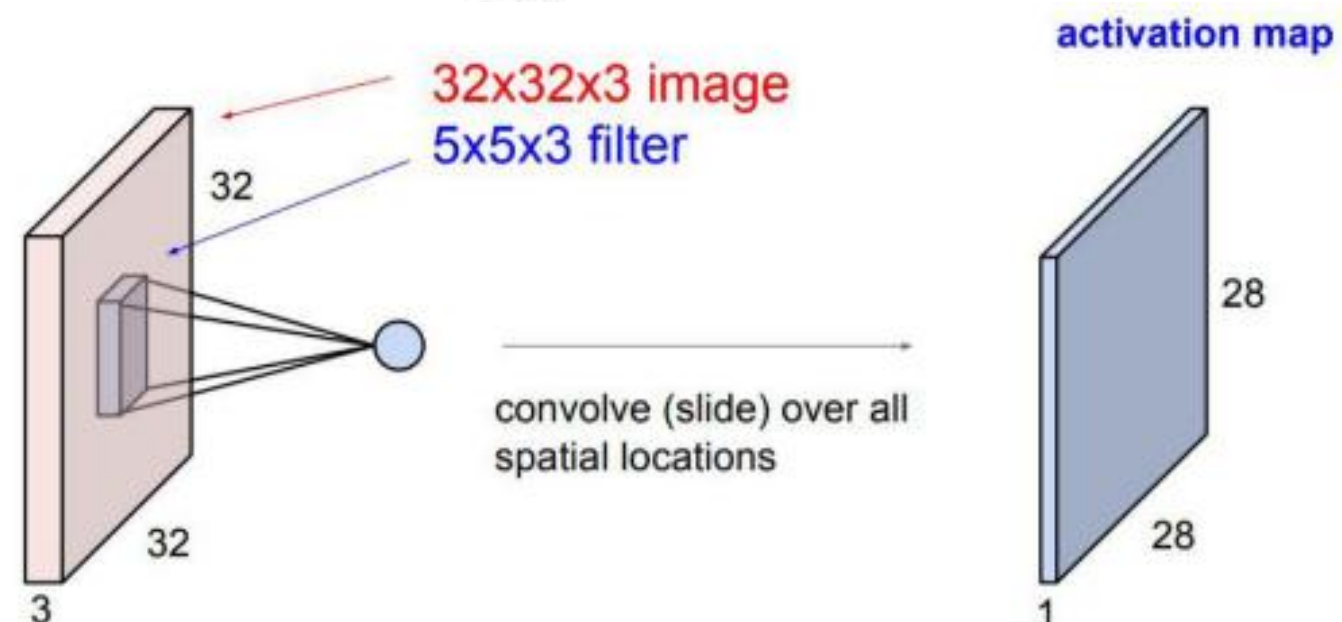
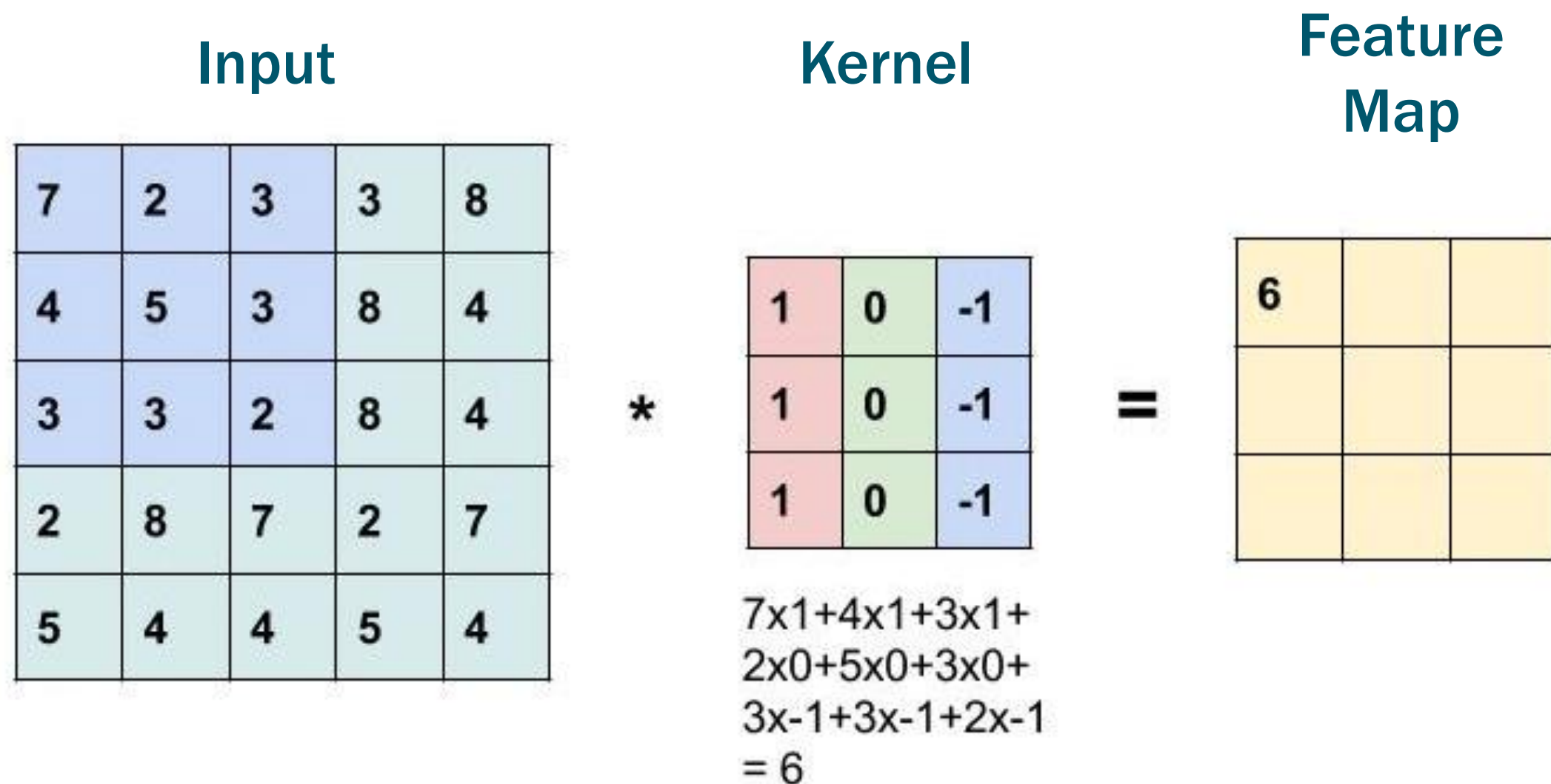
Qué son las CNN

- Las Redes Neuronales Convolucionales (CNN) disponen de una serie de capas especiales que realizan operaciones matemáticas conocidas como CONVOLUCIÓN.
- Una convolución es una operación matemática que combina dos funciones para producir una tercera.
- En las CNN se combinan los datos con un kernel (filtro), para dar una nueva matriz.

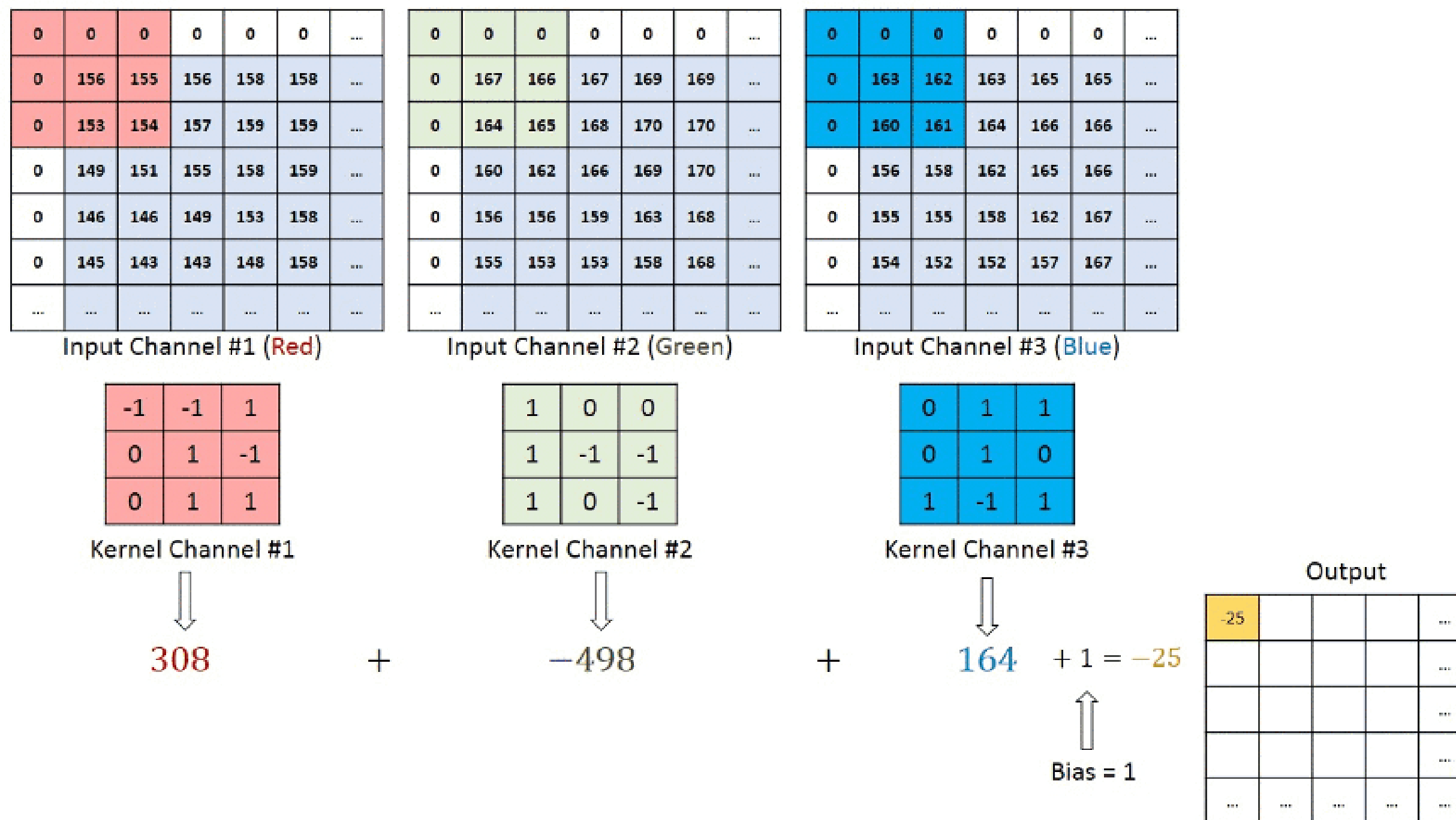
Qué son las CNN



Qué son las CNN



Qué son las CNN



Tipos de Kernels



Tipos de Kernels



$$S_x = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix}$$



$$S_y = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{pmatrix}$$



$$S = \sqrt{S_x^2 + S_y^2}$$



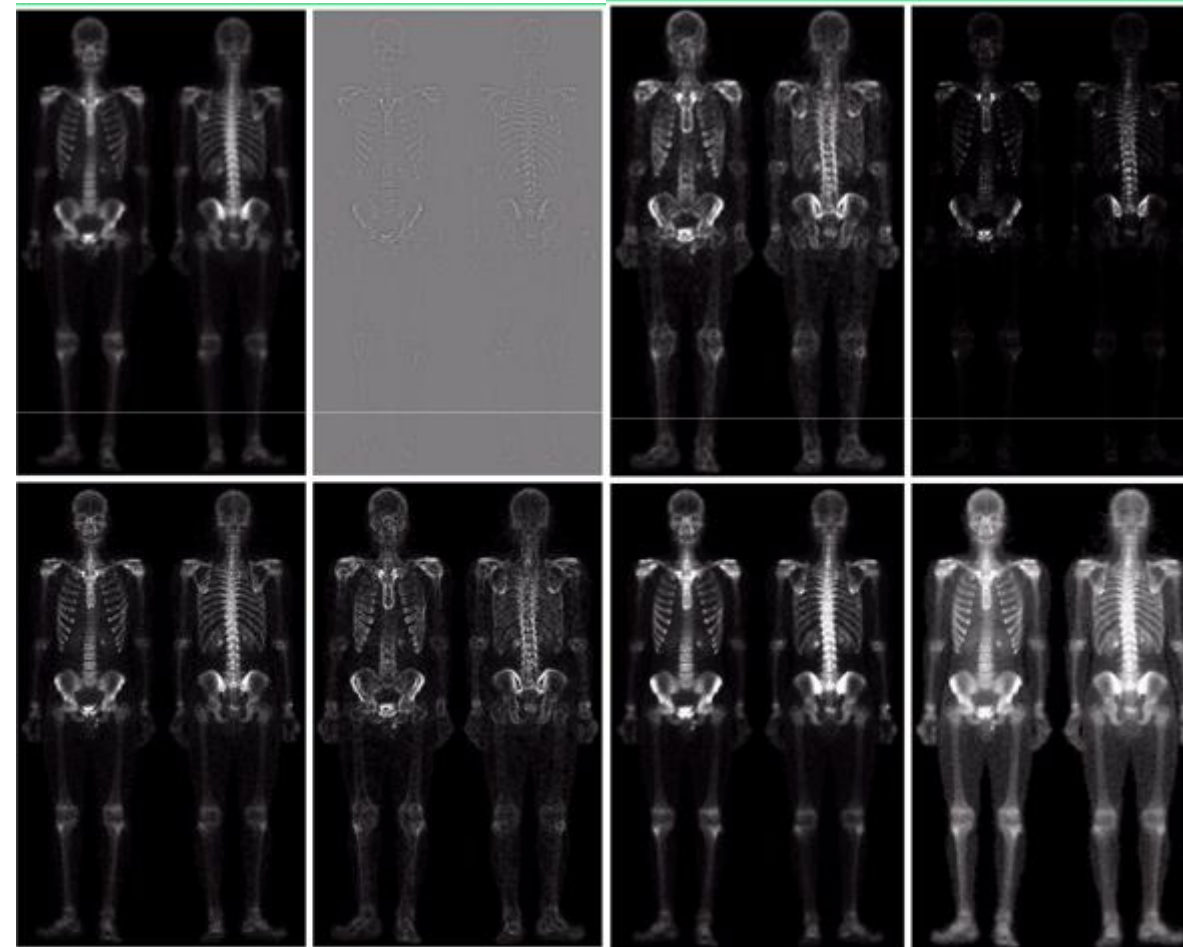
$$P_x = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}$$



$$P_y = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{pmatrix}$$



$$P = \sqrt{P_x^2 + P_y^2}$$

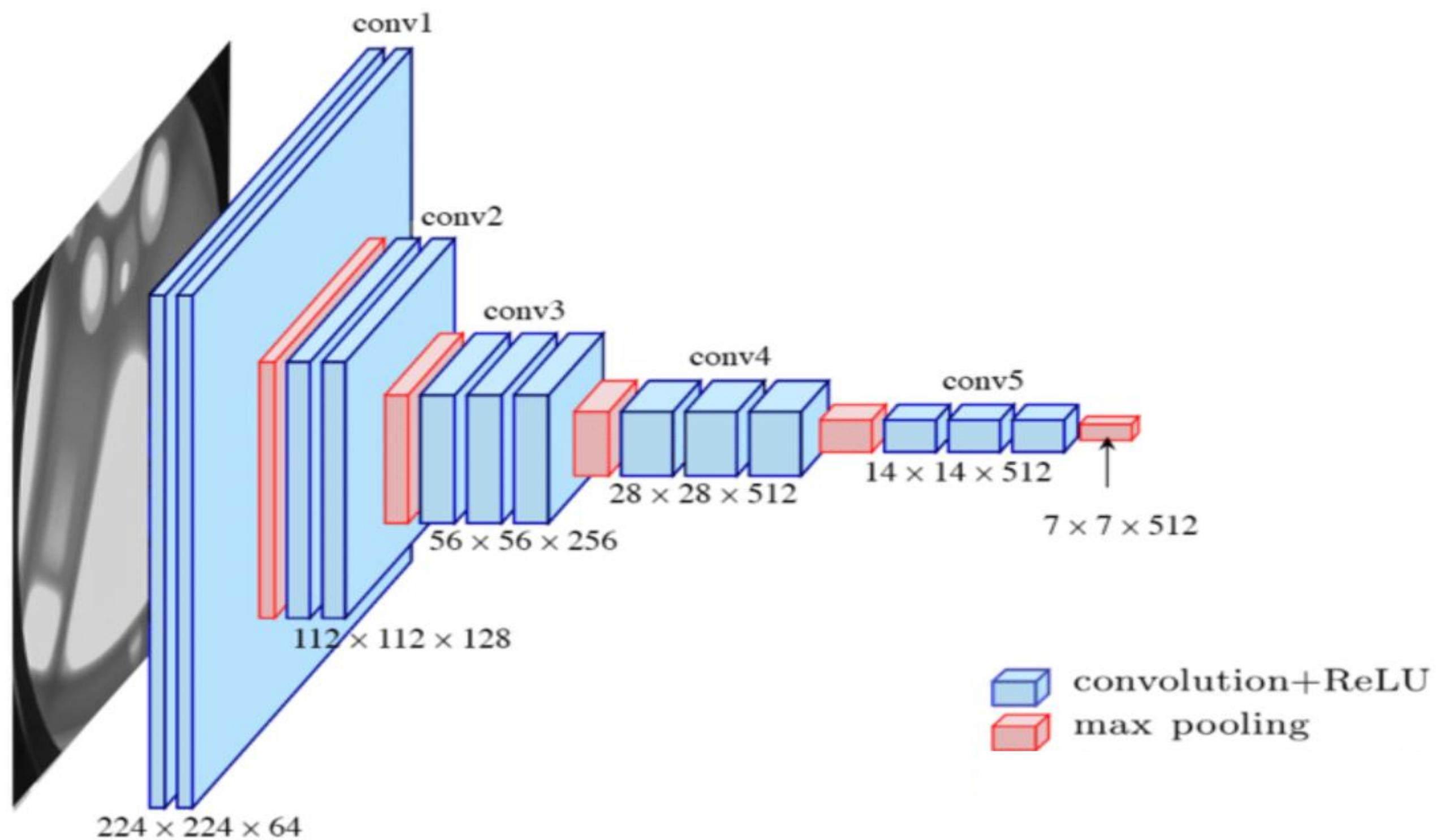


Ejemplo: <https://setosa.io/ev/image-kernels/>

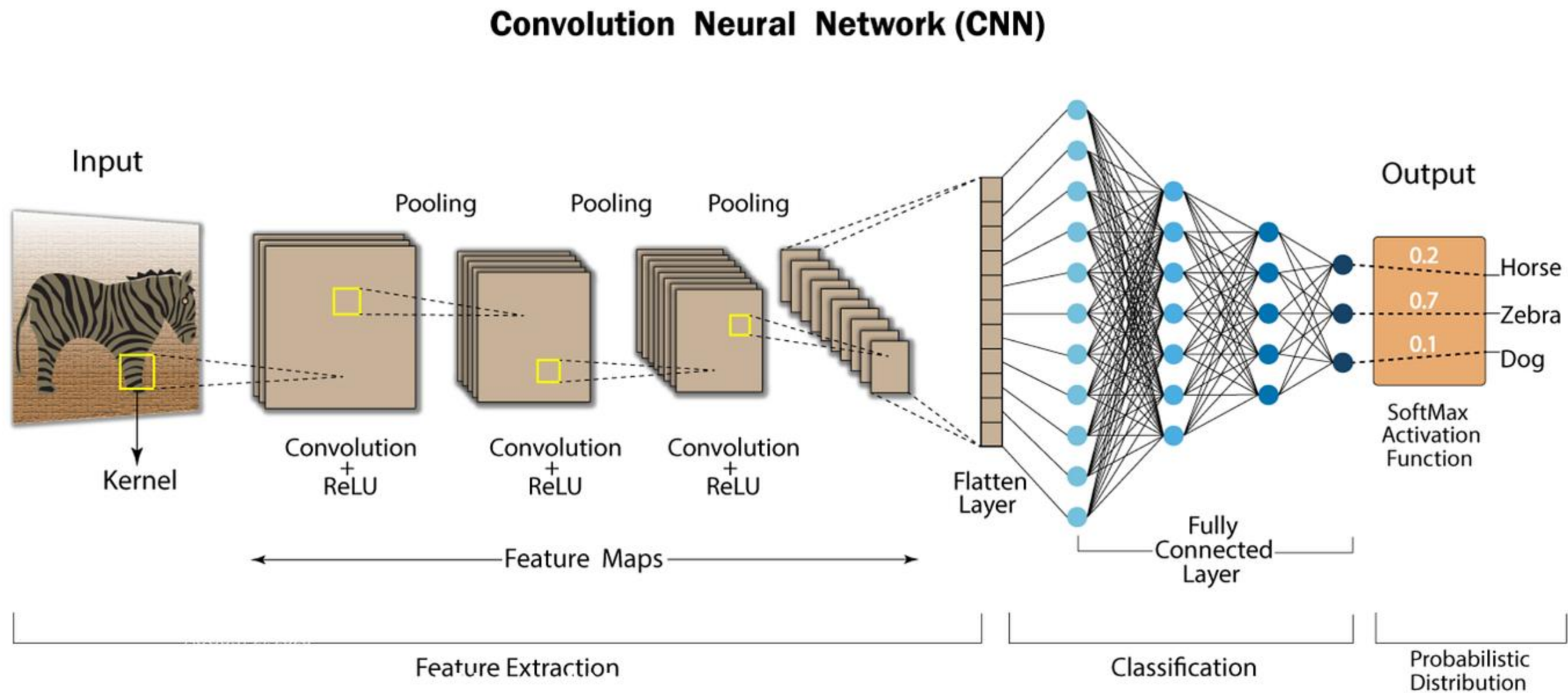
04

Arquitectura de una CNN

Arquitectura



Tipos de Kernels



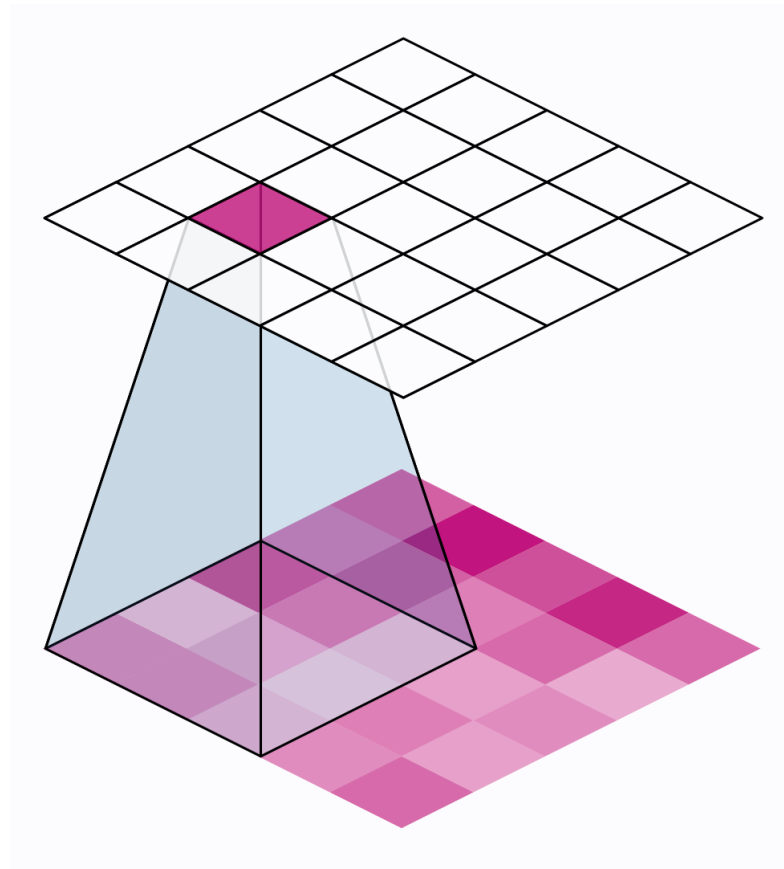
05

Aspectos de las CNN

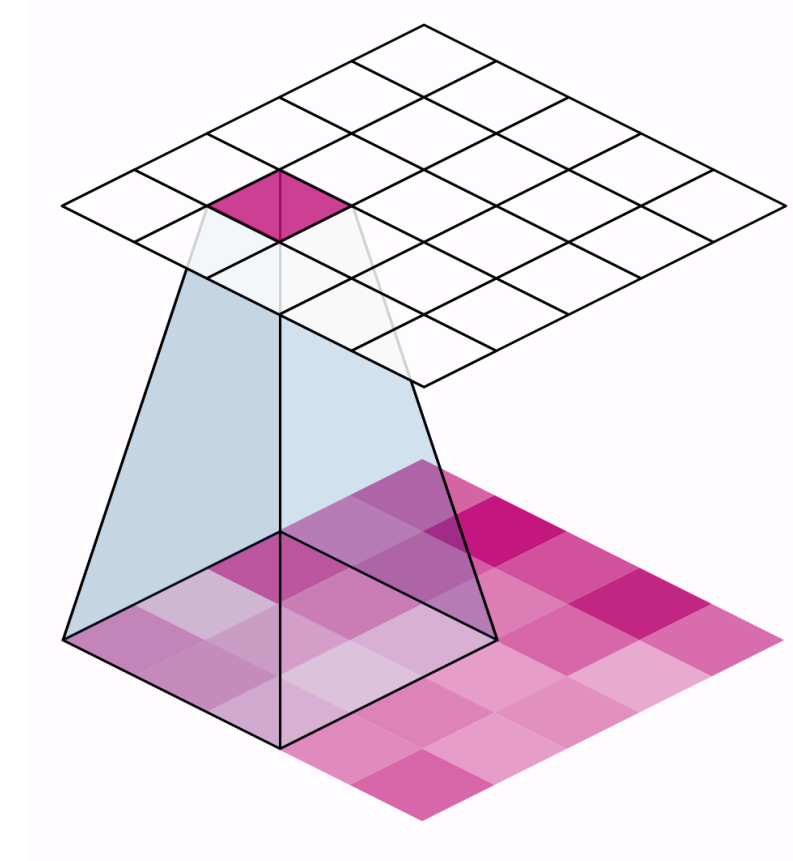
Stride

Hiperparámetro que controla cómo se desplazan los filtros a través de la imagen.

Stride = 1

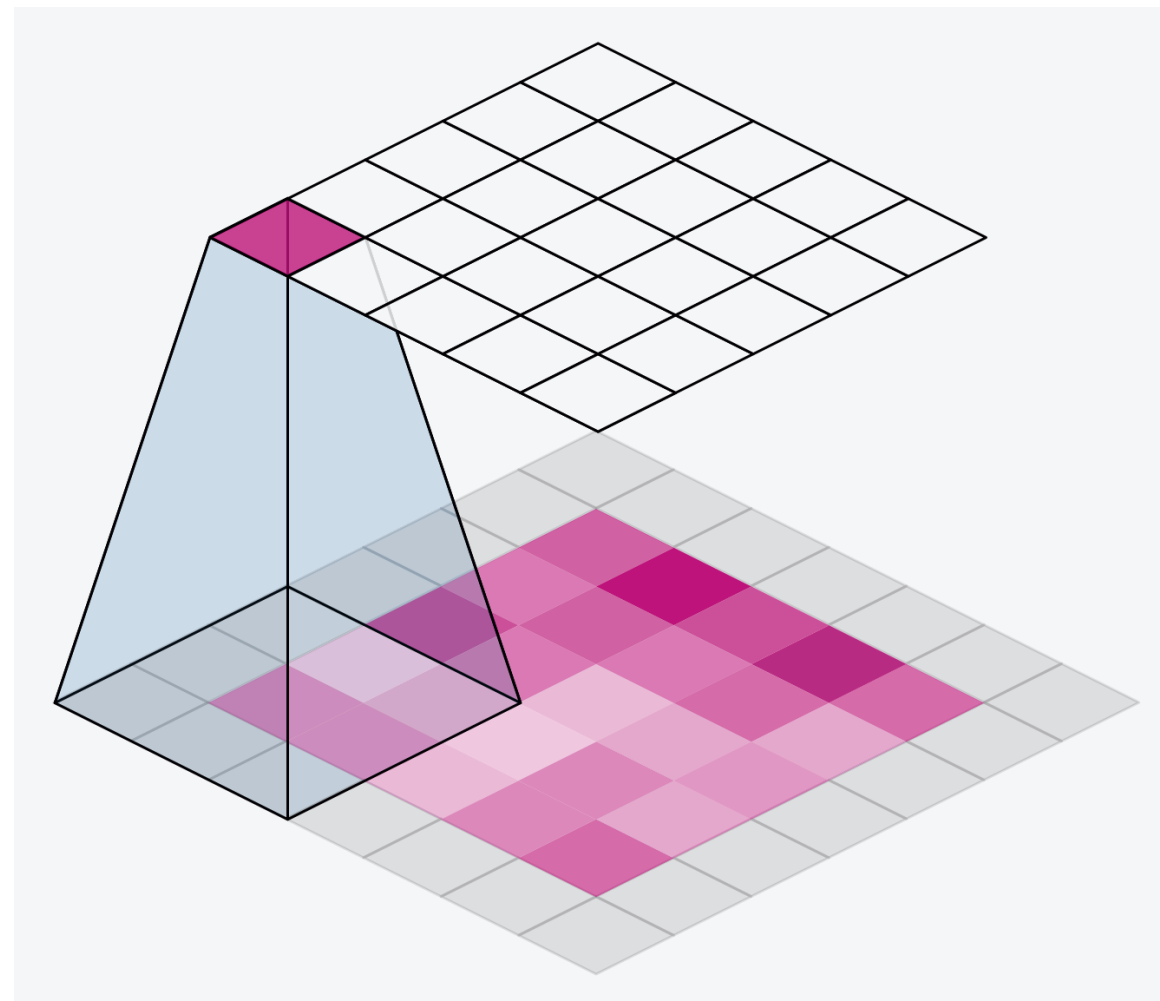


Stride = 2

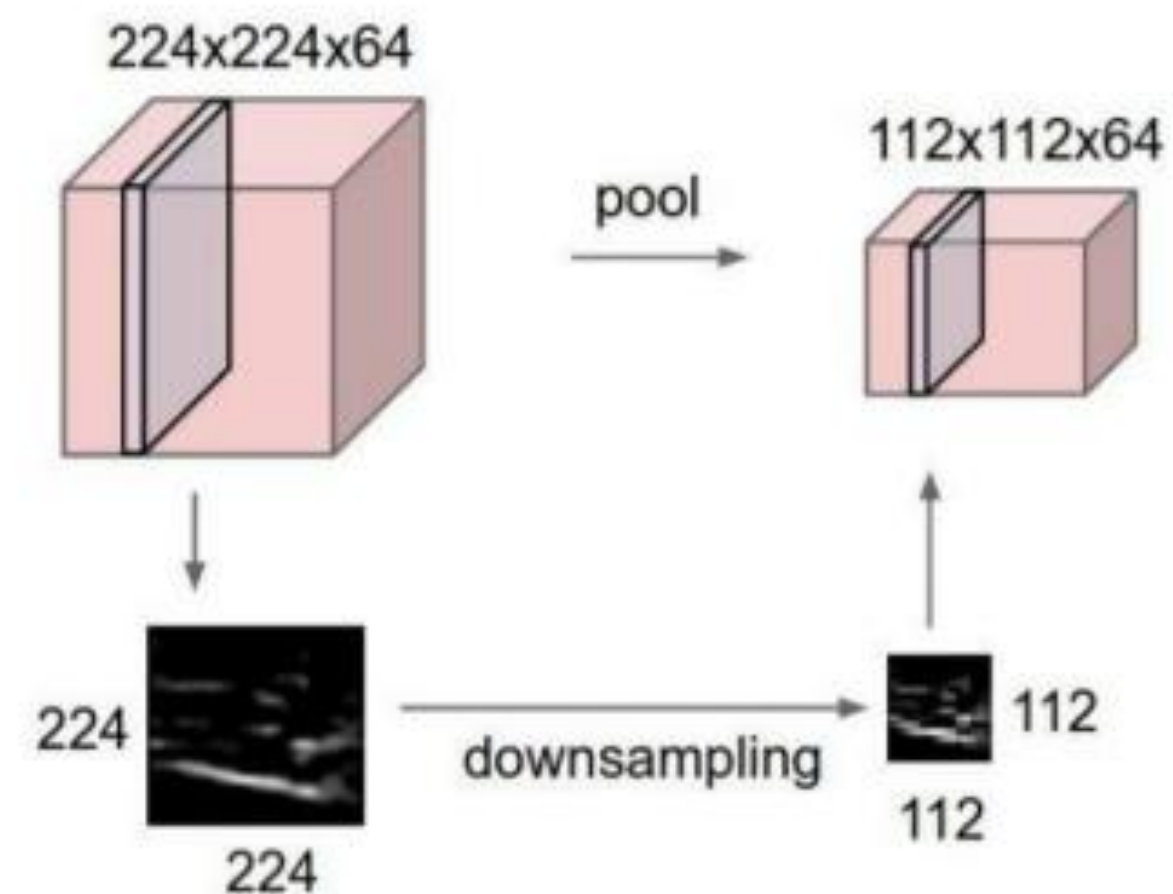
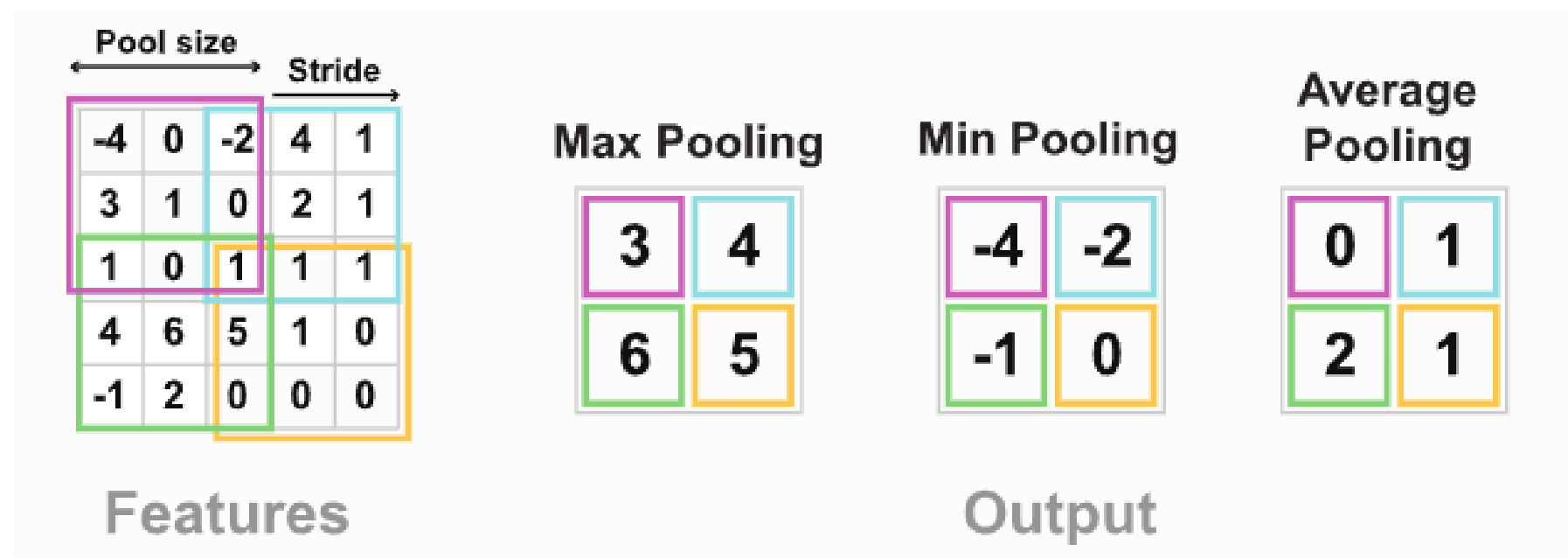


| Pading

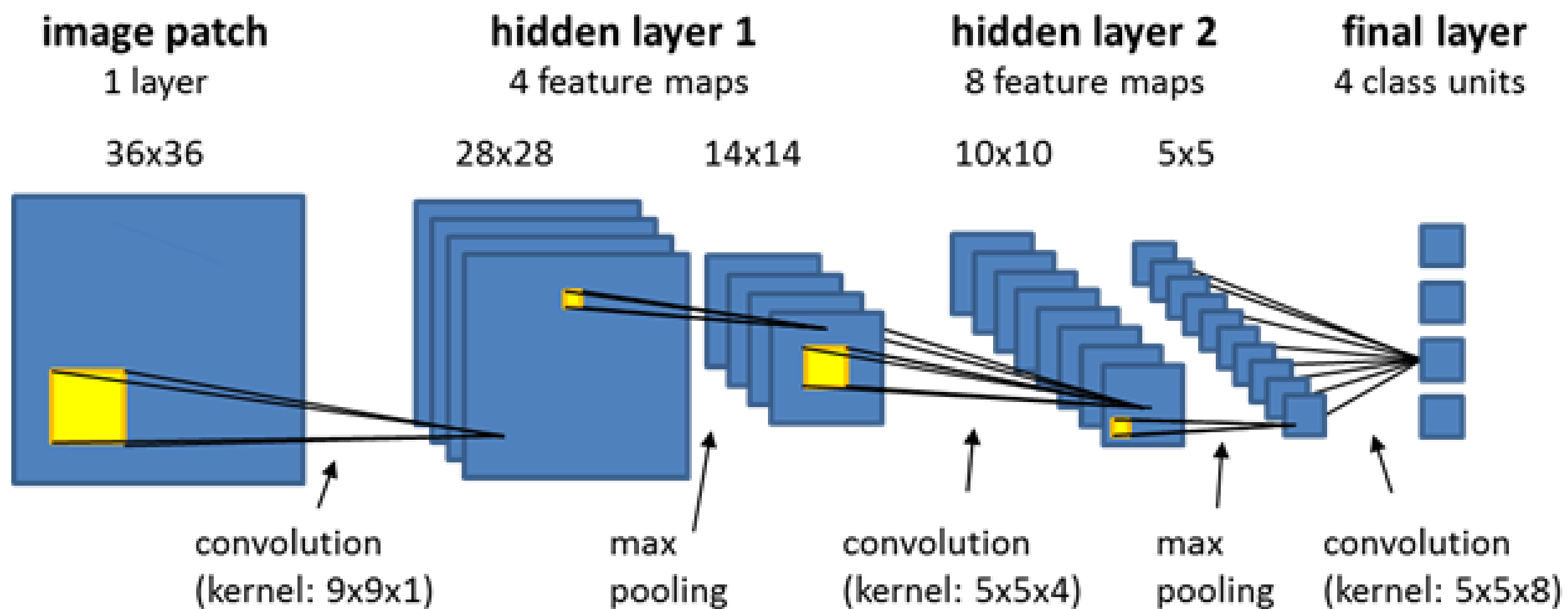
Se pueden “enmarcar” los datos a ceros



Pooling layers



Resumiendo



「muchas gracias.」